

PLSQL Exercise_3

Schema created :

```
CREATE TABLE Transactions (  
    TransactionID NUMBER PRIMARY KEY,  
    AccountID NUMBER,  
    TransactionDate DATE,  
    Amount NUMBER,  
    TransactionType VARCHAR2(10),  
    FOREIGN KEY (AccountID) REFERENCES Accounts(AccountID)  
);
```

```
INSERT INTO Transactions (TransactionID, AccountID, TransactionDate, Amount,  
TransactionType)  
VALUES (1, 1, SYSDATE, 200, 'Deposit');
```

```
INSERT INTO Transactions (TransactionID, AccountID, TransactionDate, Amount,  
TransactionType)  
VALUES (2, 2, SYSDATE, 300, 'Withdrawal');
```

Scenario 1: Monthly interest

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest  
IS  
BEGIN  
    UPDATE Accounts  
    SET Balance = Balance + (Balance * 0.01)  
    WHERE AccountType = 'Savings';  
  
    COMMIT;  
END;  
/
```

Scenario 2: Employee bonus

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus (  
    p_dept_id IN Employees.DepartmentID%TYPE,  
    p_bonus_percent IN NUMBER  
)  
IS
```

```

BEGIN
    UPDATE Employees
    SET Salary = Salary + (Salary * p_bonus_percent / 100)
    WHERE DepartmentID = p_dept_id;

    COMMIT;
END;
/

```

Scenario 3: Fund transfer

```

CREATE OR REPLACE PROCEDURE TransferFunds (
    p_source_account IN Accounts.AccountID%TYPE,
    p_target_account IN Accounts.AccountID%TYPE,
    p_amount         IN NUMBER
)
IS
    v_balance Accounts.Balance%TYPE;
BEGIN
    SELECT Balance
    INTO v_balance
    FROM Accounts
    WHERE AccountID = p_source_account
    FOR UPDATE;

    IF v_balance >= p_amount THEN
        UPDATE Accounts
        SET Balance = Balance - p_amount
        WHERE AccountID = p_source_account;

        UPDATE Accounts
        SET Balance = Balance + p_amount
        WHERE AccountID = p_target_account;

        INSERT INTO Transactions (TransactionID, AccountID, TransactionDate, Amount,
TransactionType)
        VALUES (TransactionID_SEQ.NEXTVAL, p_source_account, SYSDATE, p_amount,
'Withdrawal');

        INSERT INTO Transactions (TransactionID, AccountID, TransactionDate, Amount,
TransactionType)
        VALUES (TransactionID_SEQ.NEXTVAL, p_target_account, SYSDATE, p_amount,
'Deposit');

```

```
        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Transfer completed successfully.');
```

ELSE

```
        DBMS_OUTPUT.PUT_LINE('Insufficient balance in source account.');
```

END IF;

```
END;
```

/